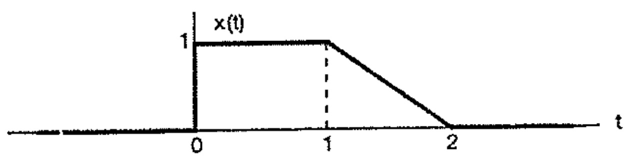
Date: 22.09.2023 Roll No.________________

III SEMESTER B.Tech (ECE, EIOT)

MID-SEMESTER EXAMINATION 2023

Course Code : ECECC05 / EIECC05 **Course Title : Signals & Systems**
Time: 1:30 Hrs **Max Marks: 25**
Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.		Questions	Marks	CO
Q 1	a	Determine whether the following discrete-time signal is periodic or not. If the signal is periodic, determine its fundamental time period. $x[n] = 2exp\left(\frac{-j\pi n}{4}\right) + exp\left(\frac{j3\pi n}{8}\right) - 2exp\left(\frac{-j5\pi n}{4}\right) + exp\left(\frac{j7\pi n}{8}\right)$. Is this a power or energy signal?	04	CO1
	b	Given the signal x(t) as shown in following Figure. Sketch $x\left(1-\frac{3}{2}t\right)$. 	01	CO1
Q 2	a	Determine whether the system described by $y(t) = \int_t^{t+1} x(\tau - a)d\tau$ (where 'a' is a constant) is i. Stable ii. Causal	01	CO1
	b	Given that $y[n] = \sum_{k=-5}^{5} 2^k x[n-k]$. Find the impulse response of the system. Is the system LTI? Is this system stable and causal?	04	CO1
Q 3	a	Give one example for each, if possible: i. One of two LTI systems is unstable but their composite is stable. ii. Two systems are time variant but their composite is time invariant. iii. One of two LTI systems is non-causal and their composite is causal. iv. Two LTI systems are unstable but their composite is stable.	04	CO2

	b	Express $\frac{1}{2}\int_{-\infty}^{\infty} x(\tau)\,[\delta(\tau-2)+\delta(\tau+2)]d\tau$ in terms of x(t)	01	CO1
Q 4	a	The unit step response of an LTI system is $y(t) = (1-e^{-2t})u(t)$. What is the response to an input of $x(t) = 2u(t) - 4u(t-1)$?	04	CO2
	b	Given that $z(t) = x(t) * y(t)$, express $p(t) = x(3t) * y(3t-6)$ in terms of z(t).	01	CO2
Q 5	a	Consider a discrete-time LTI system with impulse response $h(n) = \left(\frac{1}{2}\right)^n u(n)$. Determine the response to the following input $x(n) = (-1)^n e^{j\pi n}$ for all n. Is the system stable, Causal?	04	CO2
	b	A system is excited by a discrete-time unit-ramp $(nu[n])$ function and the response is unbounded. From these facts alone, it is impossible to determine whether the system is BIBO stable or not. Why?	01	CO2

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